

CSE 4050 - Web Application Development
Project Proposal (9/13/26)
Dr. Ghazi

Library Management System
"Pocket Librarian"

Team Lead: Michael Donnelly (009191540@coyote.csusb.edu)
Owen Latrasse (008861860@coyote.csusb.edu)
Jacob Otero (007576628@coyote.csusb.edu)
Jack Sadler (007426465@coyote.csusb.edu)

1. Project Title • A clear, descriptive name for the web application.

Pocket Librarian: A Web-Based Library Reservation and Management System

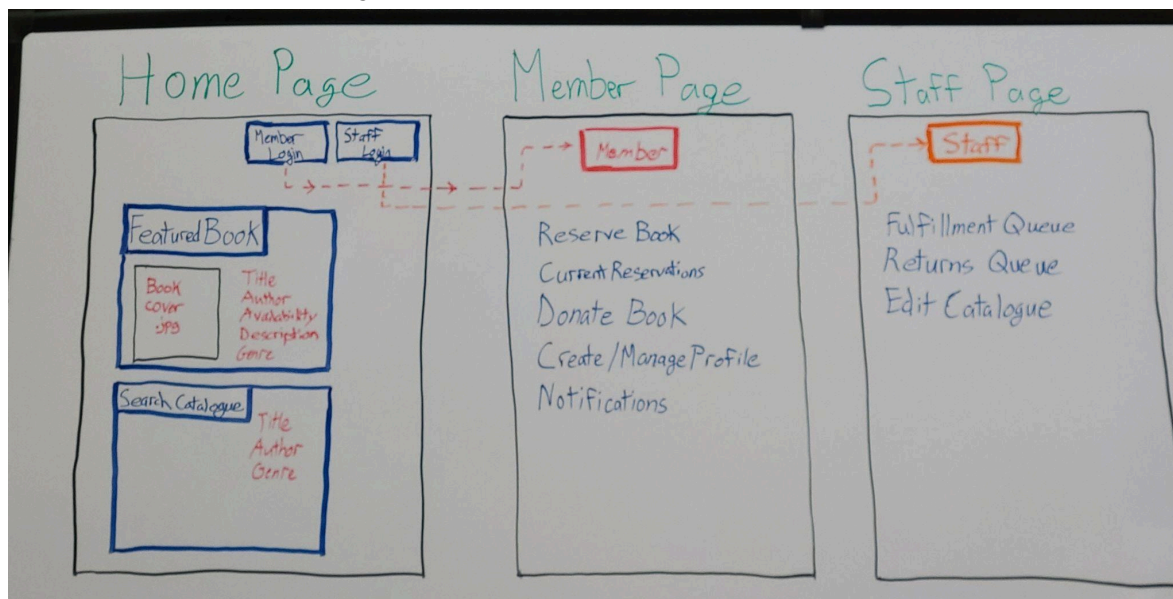
2. Problem Definition

The application addresses the problem of library members having to wait in line to request and checkout books. Library Members experience this problem because long or inconvenient wait times may discourage them from using library services as often. A web application is appropriate because it allows customers to skip lines and order books from the convenience of home. Instead of managing a physical line, library staff can instead focus on getting books ready for easy pickup by members.

3. Goals and Objectives List 3–5 concrete goals.

1. Allow members and staff to create and manage accounts with role-specific access.
2. Allow members to search the book catalogue, reserve books, and manage their reservations.
3. Allow library staff to add and manage book information in the catalogue database.
4. Provide library staff with a fulfillment queue showing which books need to be prepared for which members.
5. Automate a previously manual process: A physical line is being replaced by a web-based reservation and pickup system that works on desktop and mobile devices.

4. Proposed Solution / Key Features



1. Role-Based User Authentication - Member login and Staff login
2. Featured Book on Home Page - Feature will display a featured book on the Home Page. It will show an image of the book cover + title + author + availability + description + genre. The featured book will rotate to other books periodically.
3. Catalogue Search and Filtering - Search or filter book collection by title, author, genre, or other criteria.

4. Reservation Management - Allow members to create and manage their profile and account information.
5. Book Donation Requests - Allow members to submit books they would like to donate.
6. Fulfillment Queue - Any books that get reserved will enter the Fulfillment Queue for staff to find the books and set them aside for members to come and pick up. Once a book has been fulfilled, a notification will be sent to the member to come pick it up.
7. Member Notifications - When a book reservation has been fulfilled, the member will receive a notification to come pick it up.
8. Returns Queue - Allow staff to process returned books and update their availability / status.
9. Create / Manage Profile - Members can create a profile and manage it.
10. Catalogue Administration - Allow staff to add, edit, delete and update books in the catalogue

5. Target Users

- Library Members - Browse the catalogue, reserve books, view current reservations, manage their profiles, submit book donation requests, and receive notifications.
- Library Staff - View reservation requests, process books for pickup, process returned books, modify catalogue information, and update book availability.

6. Technical Approach

Pocket Librarian will use a client-server architecture in which the front end communicates with a backend / API connected to a SQL database.

- Front End: HTML5, CSS3, and JavaScript will be used to create the Home, Member, and Staff interfaces.
- Backend/API: The backend will process requests from the front end and handle application logic.
- Database: A SQL database will store information for members, staff, book, reservations, notifications, and donations.
- Authentication: Role-based authentication will separate Member and Staff access.
- Responsive Design: The application will be designed to work on both desktop and mobile devices.
- Testing: The application will be tested to verify that the major workflows and database operations function correctly.

7. Project Scope / Deliverables

1. A functional web-based Pocket Library application
2. Member and staff user interfaces
3. Secure user authentication
4. Searchable book catalogue
5. Reservation and fulfillment system
6. Returns processing system
7. Member notification system
8. SQL database

9. Backend/API connecting the user interface to the database
10. Responsive interface for desktop and mobile browsers
11. Application source code
12. Testing and project documentation

8. Timeline (Using Agile)

Sprint	Date	Main Goals
Sprint 0	Sep 3 - 18	Finalize requirements, UI mockups, database design, Git / project setup
Sprint 1	Sep 19 - Oct 2	Member / staff login, Featured Books, basic navigation, initial Users / Books
Sprint 2	Oct 3-16	Catalogue search / filtering, book details, catalogue database integration
Sprint 3	Oct 17-30	Reservation Management, Profile Management, Donation Requests
Sprint 4	Oct 31 - Nov 13	Fulfillment Queue, Returns Queue, Catalogue Administration
Sprint 5	Nov 14 - Nov 27	Notifications, Reservation -> Fulfillment -> Ready for pickup workflow, integration testing
Sprint 6 - Final Refinement & Deployment	Nov 28 - Early Dec	Bug fixes, usability improvements, responsive design, final testing, deployment, documentation

9. Success Criteria

1. Members can successfully create an account and login.
2. Staff and members receive the correct role-based interface.
3. Users can browse and search the catalogue.
4. Members can successfully reserve available books.
5. New reservations correctly appear in the staff fulfillment queue.
6. Staff can mark reserved books as ready for pickup.
7. Members receive a notification when a reservation becomes ready.
8. Staff can process returns and modify catalogue information.
9. Application data is correctly stored and retrieved from the database.
10. The application functions without critical errors on common web browsers.

10. Conclusion

Pocket Librarian will provide a centralized web application through which library members can search for books, manage reservations, and receive notifications while staff can process reservations, returns, and catalogue updates. The system will improve communication between members and staff by connecting these activities through a shared backend and database. The project's clearly defined member and staff roles, limited feature set, and use of standard web development technologies make it feasible to design, implement, test and deploy within the semester.